



Global Renewable Energy in 2025 How Far Have We Come?

A data-driven overview of renewables across primary energy and electricity.

For policy analysts and energy-sector
strategists
Source: Our World in Data

75% of Global GHG Emissions Still Come from Fossil Fuels

Since the Industrial Revolution, energy systems have been dominated by fossil fuels—driving climate change and major health impacts.



75%

of global greenhouse gas emissions come from burning fossil fuels for energy.



5M+

premature deaths per year from local air pollution linked to fossil fuel combustion.

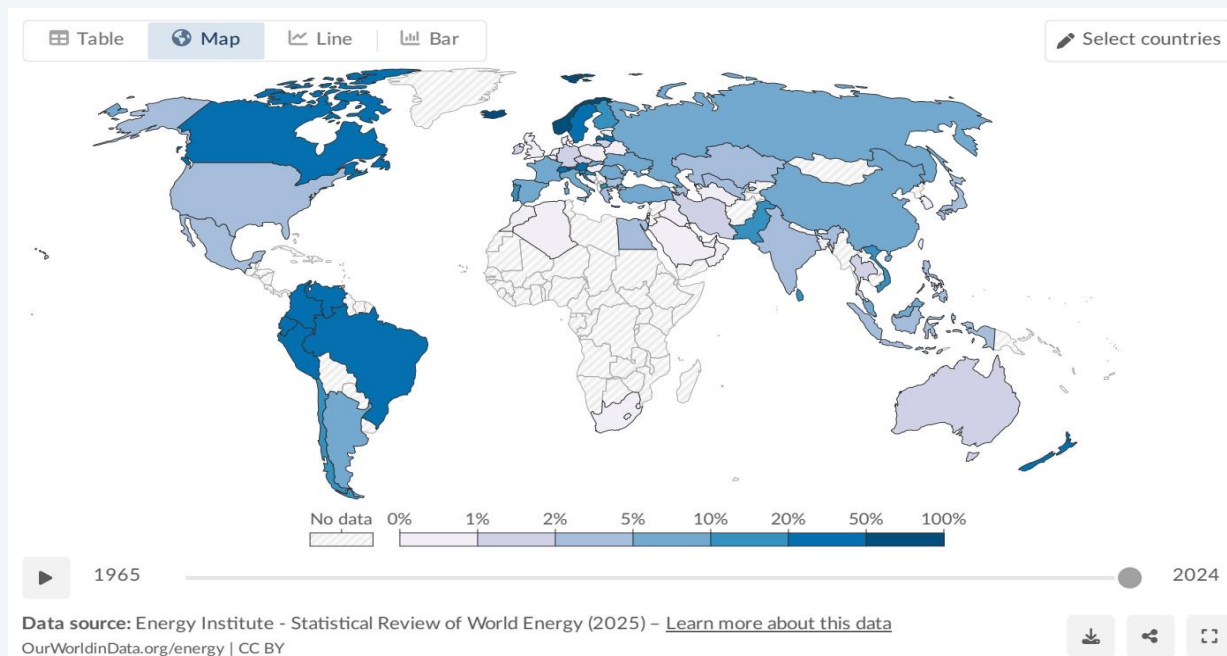


Implication for strategy

Rapidly shifting to low-carbon sources—renewables and nuclear—reduces both CO₂ emissions and harmful air pollution.

Renewables Now Supply ~14% of Global Primary Energy

Primary energy includes electricity, transport, and heating. (2024, substitution method)



Headline

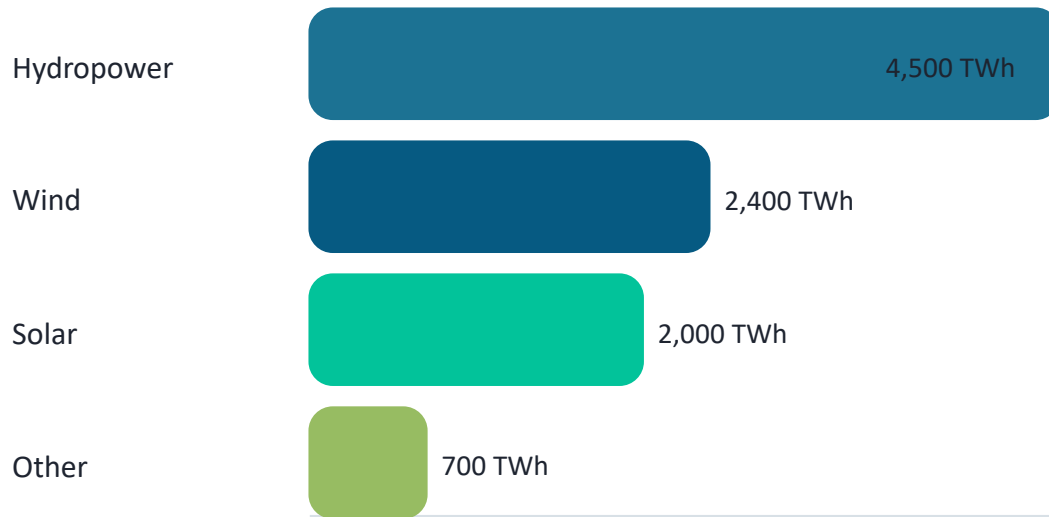
~1 in 7 units of global primary energy is renewable.

Why the share is lower than electricity:

- Transport still relies heavily on oil
- Heating still relies heavily on gas
- Electricity has seen the fastest progress

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

Approx. global renewable generation by technology (2024, TWh).

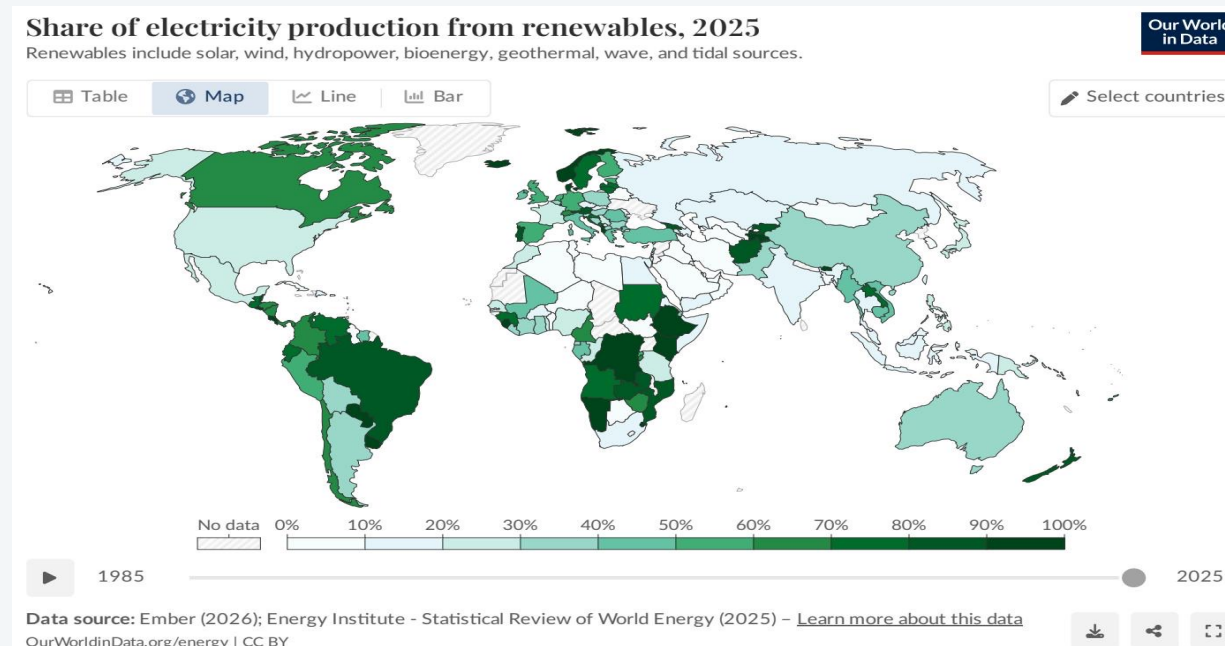


What to notice

- Wind + solar (~4,400 TWh) now nearly match hydropower output (~4,500 TWh).
- Acceleration since ~2010 is concentrated in wind and solar.
- Hydro is still the largest single source (~half of renewable electricity).

Almost One-Third of Global Electricity Now Comes from Renewables

Global renewable electricity share is ~30% (2025); wide variation by country.



Source: Our World in Data (Renewable Energy), Ember (2026), Energy Institute (2025)

Selected renewable electricity shares (approx.)

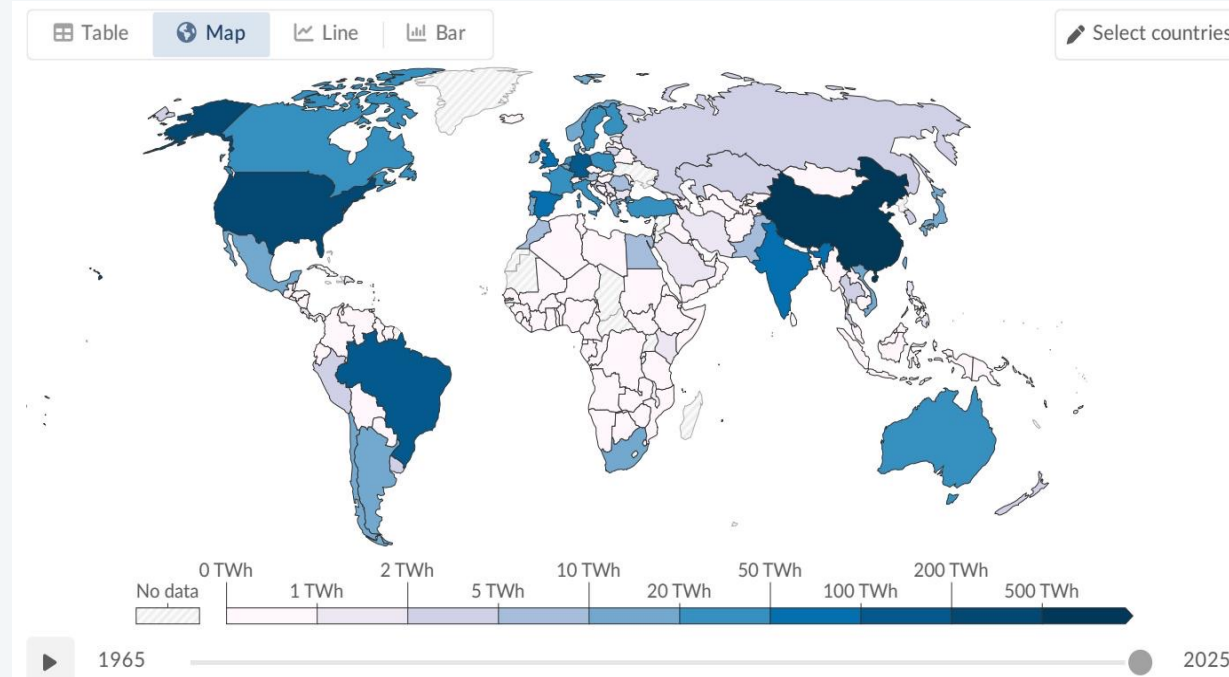
Country/Region	Renewables
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

Pattern

Hydro-rich systems (Nordics, Canada, Brazil) reach 60–98%+, while parts of the Middle East remain below 10%.

Hydropower Remains the Largest Renewable Source at ~50% of Generation

Hydropower accounts for ~47% of renewable electricity (~4,500 TWh, 2024).



Top producers

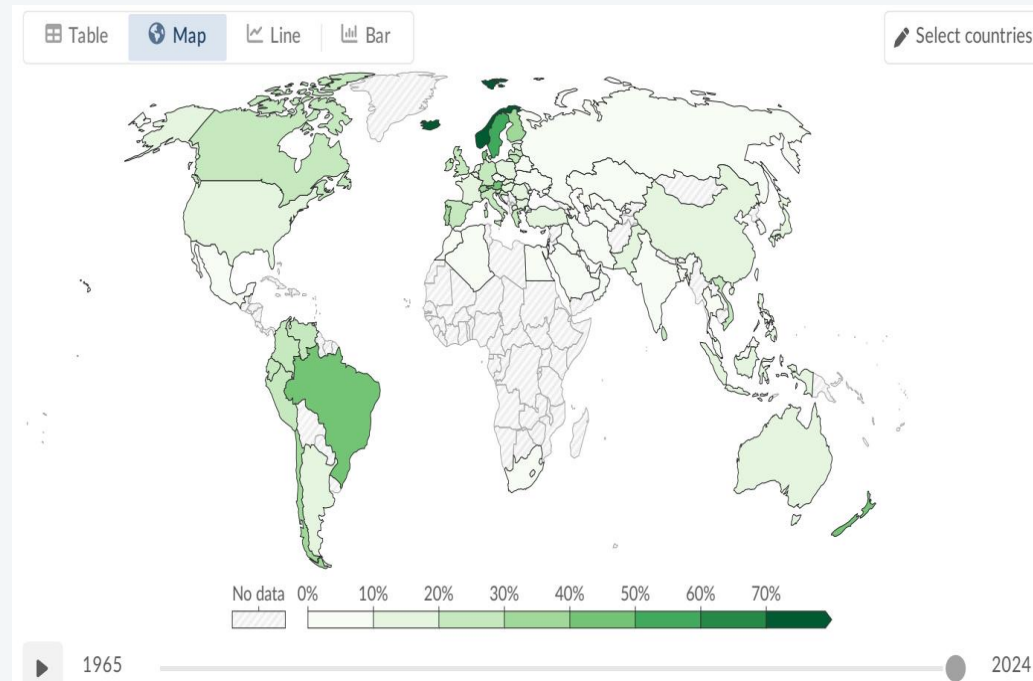
- China
- Brazil
- Canada

Strategic note

Hydro is mature and often geographically constrained—growth potential is typically lower than wind and solar.

Wind and Solar Are the Fastest-Growing Energy Technologies in History

Comparison uses generation and renewable-electricity share data from the brief (not capacity).



Wind
~2,400
TWh

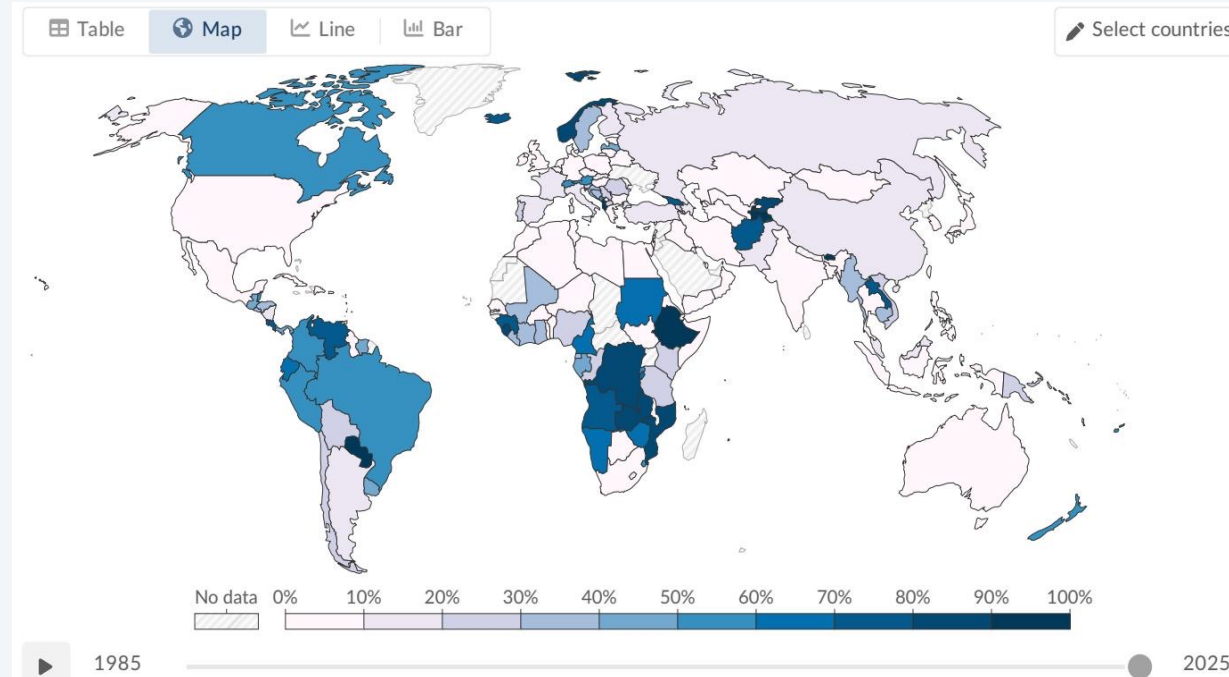
~25% of renewable
electricity

Solar
~2,000
TWh

~21% of renewable
electricity

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead

Hydropower access and deployment history shape today's renewable electricity leaders.



From the brief

- Scandinavia, Brazil, and Canada benefit from abundant hydropower (65–98% renewable electricity in examples).
- Denmark and Germany are highlighted as major investors in wind and solar.
- Many African and Middle Eastern countries remain

Key Takeaways

~14% of global primary energy (2024) and ~30% of global electricity (2025) are renewable.

Hydropower remains dominant (~47% of renewable electricity), but wind + solar now nearly match hydro output.

Wind (~2,400 TWh; ~25%) and solar (~2,000 TWh; ~21%) are expanding rapidly; solar is described as the fastest-growing since 2010.

Regional disparities persist: hydro-rich systems lead (e.g., Norway ~98%), while many countries remain below 10%.

Transport and heating lag behind electricity, keeping the overall primary-energy renewable share lower.

Forward look: accelerate clean electricity deployment and expand progress beyond power into transport and heating.